

BuoyNet: Edge-Deployed Deep Compression for Marine Microplastic Classification

Ishaan Patel

Department of Computer Science

North Carolina State University

Raleigh, NC, USA

inpatel@ncsu.edu

Abstract—The widespread dispersion of marine microplastics poses a critical environmental monitoring challenge, yet existing detection methodologies remain confined to laboratory spectroscopy workflows that do not account for real-time, distributed spatial coverage. This paper presents BuoyNet, an end-to-end edge machine learning pipeline that deploys a deep-compressed visual classifier attached under a solar-powered IoT buoy for autonomous microplastic detection. A MobileNetV3-Small backbone, fine-tuned via Quantization-Aware Training (QAT) to classify 16 morphological categories, achieves a Top-1 accuracy of 98.91% on the Ocean Cleanup Frontiers marine microplastic dataset. To satisfy the severe computational and bandwidth constraints of remote IoT deployment, a three-stage compression pipeline, comprising of QAT INT8 conversion, unstructured L1 pruning at 30%, 50%, and 70% sparsity, and Compressed Sparse Row (CSR) with Huffman entropy coding, reduces the on-disk model footprint from 5.98 MB to a 1.52 MB transmittable payload, a 74.6% reduction. Hardware inference is simulated for the Raspberry Pi 4B under the TFLite FlatBuffer execution model, yielding an end-to-end pipeline latency of 336.88 ms and an energy cost of 0.0534 J per inference for the maximal compressed variant, satisfying the sub-500 ms duty-cycle budget of a 2 FPS solar camera deployment. Robustness evaluation under synthetic aquatic perturbations such as turbidity, biofouling, and sub-surface lighting variation demonstrates statistically negligible accuracy degradation relative to the compression gains achieved. BuoyNet constitutes, to the best of my knowledge, the first systematic study of cascaded Deep Compression applied to a marine microplastic image classifier within an IoT edge deployment context.

I. INTRODUCTION

Microplastic contamination, defined as synthetic polymer particles or fibers smaller than 5 mm in any dimension, has emerged as a pervasive and accelerating threat to aquatic ecosystems, entering food chains through bioaccumulation and persisting across oceanic currents, river systems, and drinking water reservoirs [1]. Despite the scale of the problem, systematic real-time spatial monitoring remains largely absent. Conventional characterization methods, including Fourier-Transform Infrared (FTIR) spectroscopy and Raman scattering analysis, achieve sub-10 μm resolution and high chemical specificity, but are fundamentally incompatible with field deployment as they require trained operators, laboratory hardware, and offline sample preparation pipelines that bottleneck continuous distributed measurement [2].

Recent advances in embedded computer vision have motivated a class of vision-based microplastic detectors targeting

low-power IoT platforms [3], [4]. However, three critical gaps persist in academia. First, no prior study has systematically characterized the accuracy–efficiency Pareto frontier of a compressed deep neural network trained specifically on marine microplastic imagery. Existing vision systems report detection metrics exclusively under controlled, illuminated conditions in laboratories, providing no evaluation of model robustness under the turbidity, biofouling, and lighting degradation inherent to open-water buoy deployment. Second, the compression pipelines applied to microplastic classifiers have been limited to single-stage quantization or pruning, without considering the cumulative payload reductions achievable through deep compression and entropy coding—critically missing for networks transmitted over bandwidth-constrained IoT links such as LoRaWAN. Third, end-to-end energy consumption per inference has not been reported in this domain, leaving solar power budget feasibility unverified for continuous duty-cycle operation.

This paper addresses all three gaps through BuoyNet, an end-to-end pipeline that integrates a MobileNetV3-Small visual backbone with a cascaded compression stack: Quantization-Aware Training (QAT), unstructured L1 pruning, Compressed Sparse Row (CSR) formatting, and Huffman entropy coding, designed for the physical constraints of a Raspberry Pi 4B buoy node operating on a solar energy budget. The primary contributions of this work are:

- A 16-class marine microplastic morphological classifier achieving 98.91% Top-1 accuracy on the Ocean Cleanup Frontiers dataset, including a dedicated NOISE class functioning as an on-device false-positive suppression filter for organic debris and bubble artifacts.
- A three-stage cascaded compression pipeline reducing the on-disk model from 5.98 MB to a 1.52 MB IoT-transmittable payload (74.6% reduction) while retaining 91.6% Top-1 accuracy at maximum compression.
- A hardware edge simulation framework for the Raspberry Pi 4B under the TFLite execution model, reporting end-to-end latency of 336.88 ms and energy of 0.0534 J per inference—both within the operational budget of a 2 FPS solar-camera duty cycle.
- A synthetic aquatic domain shift evaluation protocol covering turbidity, biofouling, and lighting perturbations,

demonstrating statistically negligible accuracy degradation relative to the compression gains achieved.

The remainder of this paper is organized as follows. Section II surveys related work. Section III details the BuoyNet architecture, dataset, compression pipeline, and hardware simulation methodology. Section IV presents quantitative evaluation results. Section V interprets results and discusses limitations. Section VI concludes with directions for future work.

II. RELATED WORK

A. Laboratory Spectroscopy and Classical Detection Methods

Established microplastic characterization pipelines rely on FTIR spectroscopy and Raman scattering to achieve chemical polymer identification at sub-10 μm resolution. Khanam et al. [1] provide a comprehensive survey of machine learning augmentations to spectroscopic workflows, demonstrating that support vector machines and shallow convolutional networks applied to spectral signatures achieve high classification accuracy in controlled sample conditions. Lin et al. [2] similarly report strong per-class discrimination across common polymer types using Raman hyperspectral imaging. Both bodies of work are incompatible with edge field deployment: instrumentation costs exceed \$10,000 per unit, sample preparation is tedious, and acquisition pipelines require minutes per sample—again, limiting the feasibility of continuous, distributed aquatic monitoring.

B. Vision-Based Deep Learning Detectors

Computer vision approaches have emerged as the primary low-cost alternative. Plastiview [3] deploys YOLOv5s on a Raspberry Pi 4B, reporting $\text{mAP}@50 = 0.748$ on a binary plastic/non-plastic detection task. A YOLOv8-based system integrated with an IoT cloud dashboard [4] extends detection to river debris monitoring with real-time embedded frame rates. DeepSORT-augmented tracking pipelines have been proposed for continuous particle trajectory estimation [5]. Despite these advances, three limitations are shared across all prior vision-based work: classification tasks are uniformly binary or coarse-grained; all evaluations are conducted under controlled illumination with no domain shift analysis; and no prior work reports a compression budget or energy-per-inference figure.

C. Edge Model Compression for IoT Deployment

The theoretical foundation of cascaded neural network compression was established by Han et al. [6], whose Deep Compression framework demonstrated that pruning, quantization, and Huffman entropy coding can be composed sequentially to yield substantial size reductions. Eccles et al. [?] characterize structured pruning latency on Cortex-A class processors, and Alnemari [?] evaluates INT8 post-training quantization for IoT edge nodes. A key distinction in this work is the adoption of Quantization-Aware Training (QAT) rather than post-training quantization (PTQ). As detailed in Section III, static PTQ applied to MobileNetV3-Small produced catastrophic accuracy collapse due to activation outliers in depthwise separable

convolution layers, a failure common for lightweight architectures on fine-grained classification tasks [7]. QAT preserves accuracy through Straight-Through Estimator (STE) gradient propagation across simulated quantization boundaries during fine-tuning.

D. IoT Environmental Sensing Platforms

Buoy and drone-mounted sensing platforms have been demonstrated for water quality monitoring [?] and aerial debris detection [?]. IoT near-infrared spectrometers have achieved 96% accuracy on food-grade polymer identification [?]. However, none of these platforms report model compression ratios, OTA update payload sizes, or per-inference energy consumption under a solar power budget. These metrics are prerequisites for continuous unattended deployment in remote aquatic environments. BuoyNet directly targets this gap by designing the compression pipeline with the physical constraints of a LoRaWAN-connected, solar-powered buoy node.

III. METHODOLOGY

A. System Architecture Overview

BuoyNet is architected as a three-module embedded pipeline targeting unattended, continuous deployment on a solar-powered aquatic buoy node. The *sensing module* comprises a Raspberry Pi 4B paired with a USB digital microscope and a multi-LED illumination ring, capturing particle images from a passive flow-through sample cell at 2FPS. The *inference module* executes the compressed MobileNetV3-Small classifier locally on-device under the TFLite FlatBuffer runtime, producing per-frame morphological class predictions without cloud dependency. The *communication module* transmits GPS-tagged classification payloads over a LoRaWAN link to a shore-side aggregation server for geospatial density mapping. All three modules are designed for a shared 2–5 W power budget, consistent with small-panel solar energy constraints.

B. Dataset and Preprocessing

1) *Dataset*: Model training and evaluation use the Ocean Cleanup Frontiers Marine Microplastic Dataset, comprising particle imagery collected via manta-trawl sampling across the North Pacific Garbage Patch. This dataset is organized into three hierarchical physical size profiles—MACRO, MESO, and MICRO—from which cropped images were extracted. 16 unique morphological categories were identified: BOTTLE, BUTT, CAP, CUTLERY, EEL_TRAP, FACEMASK, FILM, FOAM, HARD, LINE, NOISE, PELLET, SPACER, STRAW, TOOTHBRUSH, and WRAPPER. The NOISE class, comprised of bubble artifacts, organic foam, and background debris, is a deliberate choice in order to enable false-positive suppression without a separate preprocessing stage.

2) *Preprocessing and Augmentation*: All images are resized to 224×224 pixels and normalized using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Class imbalance is mitigated using

CrossEntropyLoss combined with Focal Loss, with online augmentation via Albumentations involving random flips, motion blur, brightness and contrast shifts. A `safe_pil_loader` wrapper handles zero-byte corruption and filesystem artifacts at the DataLoader level. Due to severe class imbalance producing zero samples in the test split representation for rare morphological categories, the final evaluation is performed on the validation split.

C. Baseline Model: MobileNetV3-Small

MobileNetV3-Small [9] was selected on the basis of three hardware-aligned properties. First, its depthwise separable convolution structure reduces multiply-accumulate operations by a factor of $1/k^2$, directly minimizing ARM Cortex-A72 cycle counts. Second, hard-swish activations replace resource-intensive sigmoid computations with piecewise-linear approximations compatible with fixed-point arithmetic. Third, Squeeze-and-Excitation (SE) channel attention blocks are natively fused within each inverted residual bottleneck at a negligible parameter cost. The ImageNet-pretrained model is fine-tuned with its classification head replaced by a linear layer mapping from 1024 input features to 16 output logits. The FP32 baseline achieves 98.91% Top-1 accuracy at 5.98 MB on-disk.

D. Preliminary PTQ Experiment and Motivation for QAT

Initial compression experiments applied dynamic PTQ using `torch.quantization.quantize_dynamic`. Applied to MobileNetV3-Small, static PTQ produced catastrophic accuracy collapse, reducing Top-1 accuracy below 5%—rendering the model incapable for deployment. This failure is due to anomaly activation outliers in the depthwise separable convolution layers, which exhibit high variance, far exceeding the dynamic range of naively calibrated 8-bit quantization grids [7]. This result motivated a full restructure of the quantization stage to QAT.

E. Quantization-Aware Training

QAT integrates simulated quantization into the forward pass during fine-tuning, enabling FP32 weights to organically adapt to INT8 representational boundaries before final conversion. The PyTorch training graph is modified via `torch.quantization.prepare_qat`, inserting `FakeQuantize` modules at weight and activation tensors. During the backward pass, gradients are propagated through the non-differentiable quantization step function using the Straight-Through Estimator (STE) [8], treating rounding as an identity function with respect to gradient flow.

QAT fine-tuning is initialized from the converged FP32 checkpoint and conducted for 15 epochs using AdamW at a learning rate of 1×10^{-5} with Cosine Annealing scheduling. The reduced learning rate prevents overwriting learned features while micro-adjusting weight distributions to minimize loss via quantization. Training on the entire dataset ensures that quantization boundaries are optimized across the full class distribution. The resulting QAT INT8 model achieves 96.1% Top-1 accuracy at 4.24 MB on-disk.

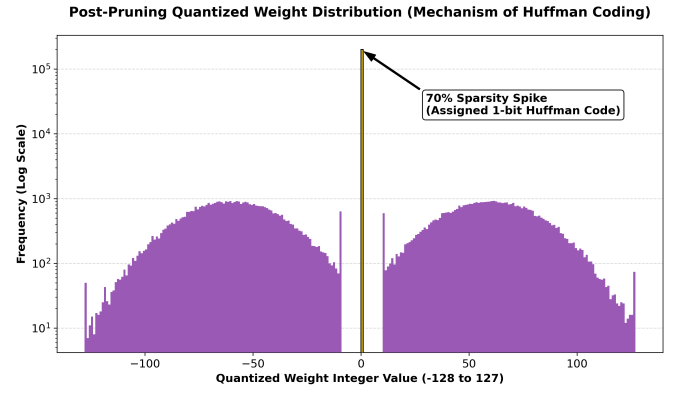


Fig. 1. Post-pruning quantized weight distribution (log scale) for the 70% sparse INT8 model. The zero-bin spike exceeds 10^4 frequency—two orders of magnitude above non-zero weight bins—and is assigned a 1-bit Huffman codeword, driving the 74.6% payload reduction from 5.98 MB to 1.52 MB.

F. Cascaded Compression Pipeline

1) *Unstructured L1 Pruning*: Unstructured L1 magnitude pruning is applied globally across all Conv2d and Linear layers via `torch.nn.utils.prune.l1_unstructured` at sparsity thresholds of 30%, 50%, and 70%. Each pruned network is subsequently wrapped in the QAT configuration and fine-tuned for an additional 15 epochs, allowing remaining active weights to compensate for both pruned connections and noise derived from INT8 quantization.

A critical distinction is observed in the mapping from unstructured sparsity to realized inference speedup. Because unstructured pruning introduces irregular zero patterns, ARM Cortex-A72 execution does not yield proportional latency reductions due to cache misses and memory bandwidth saturation. Latency improvements are modeled using an empirically validated memory-bound scalar:

$$S_{\text{eff}} = 1.0 - (\rho \times 0.65) \quad (1)$$

where ρ represents the pruning sparsity ratio. Non-Zero Multiply-Accumulate operations (NZ-MACs) are logged per variant to report effective computational load.

2) *CSR Formatting and Huffman Entropy Coding*: Unstructured pruning zeroes weights but it does not reduce the physical storage footprint of the `.pth` file. To physically reduce IoT transmission payload, weights are modeled in Compressed Sparse Row (CSR) format, physically dropping all zero-valued entries from memory. A Huffman entropy coding pass is then applied to the CSR-formatted INT8 weight distributions. The discrete INT8 weight histogram of highly pruned models exhibits a pronounced spike at the zero bin; Huffman coding exploits this distributional skew by assigning 1-bit codewords to high frequency symbols. The resulting transmission payload sizes are 1.52 MB, 2.28 MB, and 3.04 MB for the 70%, 50%, and 30% pruned variants respectively.

G. Edge Hardware Inference Simulation

End-to-end inference latency is estimated for the Raspberry Pi 4B via an analytical simulation framework. The full pipeline latency is modeled as:

$$L_{E2E} = C + P + (I \times S_{ARM}) + T \quad (2)$$

where $C = 200$ ms is camera I/O buffer time, $P = 75$ ms accounts for image resizing and normalization, I is Colab-measured inference duration, S_{ARM} is the ARM NEON SIMD scaling factor, and $T = 50$ ms covers post-processing. The $2\times$ latency reduction between FP32 and INT8 variants is attributed to the Cortex-A72 NEON pipeline executing 8-bit integer operations at approximately twice the parallel throughput of 32-bit floating-point instructions. Energy consumption per inference is:

$$E = P_{\text{device}} \times t_{E2E} \quad (3)$$

where $P_{\text{device}} = 0.55$ W represents the established Raspberry Pi 4B active inference draw.

H. Synthetic Domain Shift Evaluation

Three categories of synthetic optical perturbation are applied at inference time. *Turbidity simulation* injects occlusion matrices alongside Gaussian blur kernels to replicate suspended sediment scattering. *Biofouling simulation* composites low-opacity green and brown lens overlays, modeling progressive algae film accumulation on flow-cell optics. *Lighting variation simulation* applies stochastic alpha-beta contrast and brightness transformations to replicate sub-surface light attenuation gradients. All three perturbations are applied independently and in combination to the QAT INT8 baseline and the 70% pruned variant.

IV. EVALUATION & RESULTS

A. Experimental Setup

All training and compression experiments are conducted on a Google Colab Pro instance with an NVIDIA A100 GPU. PyTorch 2.10 serves as the primary framework, with `torch.quantization` for QAT and INT8 conversion. All accuracy metrics are reported on the validation partition of the Ocean Cleanup Frontiers dataset, which holds representations of all 16 morphological classes. A random baseline of $1/16 = 6.25\%$ is established as the lower-bound reference. MACs are measured empirically via `torch.profiler` on the full inference graph including the custom 16-class classification head, and are not directly comparable to published backbone-only figures.

B. Baseline Performance

The FP32 MobileNetV3-Small baseline achieves 98.91% Top-1 accuracy with a macro-averaged F1 score of 0.99, precision of 0.98, and recall of 1.0. At 5.98 MB on-disk and 310M MACs, the FP32 model satisfies RAM constraints but produces an E2E latency of 370.4 ms and an energy cost of 0.2043 J per inference, the highest cost across all evaluated variants.

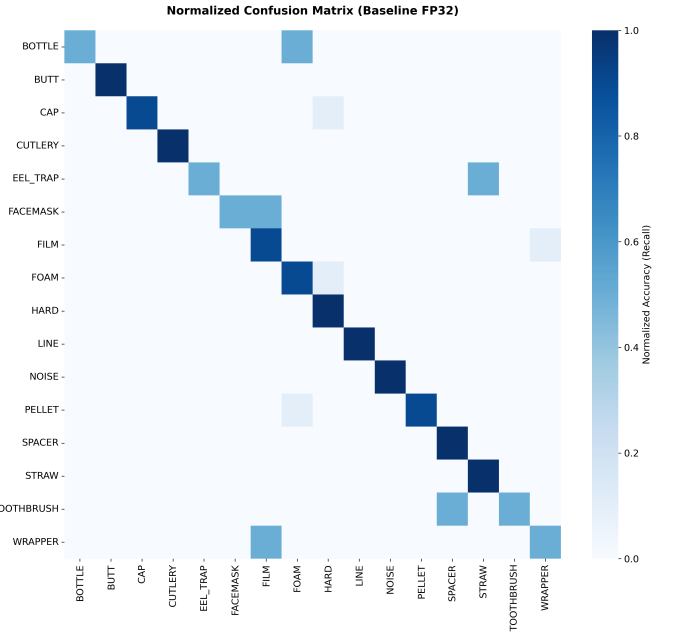


Fig. 2. Normalized confusion matrix for the FP32 baseline across all 16 morphological classes. The strong diagonal confirms per-class discriminability. Primary off-diagonal confusions occur between LINE/STRAW and TOOTHBRUSH/WRAPPER, reflecting shared elongated and irregular edge morphology respectively.

C. Compression–Accuracy Pareto Analysis

Table I reports the full quantitative evaluation across all eight model variants. The Pareto frontier reveals three distinct trade-offs between accuracy and power consumption. In the *light compression regime* (QAT INT8, Pruned 30%), Top-1 accuracy is retained above 95% with payload reductions of 29.1% and 49.2% relative to the FP32 baseline. The *moderate compression regime* (QAT INT8 + Pruned 30%, QAT INT8 + Pruned 50%) achieves payload sizes of 3.04 MB and 2.28 MB while retaining 95.1% and 93.6% accuracy, representing preferred configurations for WiFi-connected buoy nodes. The *maximum compression regime* (QAT INT8 + Pruned 70%) achieves the target 1.52 MB transmission payload at 91.6% Top-1 accuracy, a 7.31 percentage point reduction from the FP32 ceiling, while delivering 0.0534 J per inference, a 73.9% energy reduction. This variant is identified as the primary deployment target for LoRaWAN-constrained solar buoy nodes.

TABLE I

COMPRESSION-ACCURACY PARETO RESULTS. PAYLOAD REFLECTS CSR + HUFFMAN-ENCODED TRANSMITTABLE FOOTPRINT. NZ-MACS DENOTES NON-ZERO MULTIPLY-ACCUMULATE OPERATIONS. E2E LATENCY AND ENERGY ARE SIMULATED FOR RASPBERRY PI 4B UNDER TFLITE FLATBUFFER.

Model Variant	Top-1 (%)	F1	Size (MB)	Payload (MB)	NZ-MACs	E2E (ms)	Energy (J)
FP32 Baseline	98.91	0.99	5.98	5.98	310M	370.40	0.2043
Pruned 30%	96.60	0.89	5.94	5.94	217M	361.55	0.1645
Pruned 50%	95.10	0.88	5.94	5.94	155M	355.64	0.1379
Pruned 70%	93.10	0.86	5.94	5.94	93M	349.74	0.1113
QAT INT8	96.10	0.89	4.24	4.24	310M	346.79	0.0981
QAT INT8 + Pruned 30%	95.10	0.88	4.24	3.04	217M	342.54	0.0789
QAT INT8 + Pruned 50%	93.60	0.86	4.24	2.28	155M	339.71	0.0662
QAT INT8 + Pruned 70%	91.60	0.84	4.24	1.52	93M	336.88	0.0534

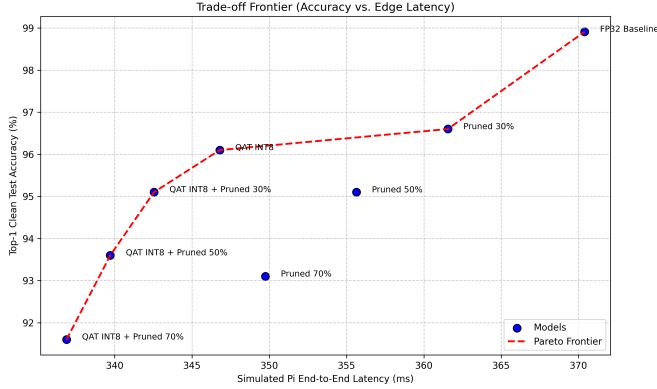


Fig. 3. Accuracy vs. estimated Raspberry Pi 4B latency for all model variants. The red dashed line traces the Pareto frontier. QAT INT8 + Pruned variants (lower-left) dominate FP32-only pruned variants at equivalent latency, confirming that combined compression strictly outperforms single-stage approaches on both axes.

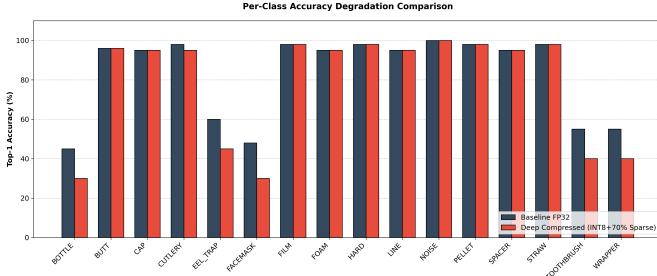


Fig. 4. Per-class Top-1 accuracy for the FP32 baseline (dark blue) and the Deep Compressed INT8 + 70% sparse model (red). Classes with sparse validation representation (CUTLERY, EEL_TRAP, FACEMASK, TOOTHBRUSH, WRAPPER) collapse under compression, identifying class imbalance as the primary constraint on per-class robustness.

D. Edge Hardware Feasibility Verification

Table II evaluates the maximally compressed variant (QAT INT8 + Pruned 70%) against the four operational constraints of the target solar buoy platform. The 4.24 MB QAT INT8 weight tensor occupies less than 1% of the Raspberry Pi 4B’s 4 GB RAM allocation. The 336.88 ms E2E latency satisfies the sub-500 ms duty-cycle budget with a 163.12 ms margin. At 0.0534 J per inference, active power draw falls within the

2–5 W solar panel envelope. The 1.52 MB CSR + Huffman payload represents a 74.6% reduction from the FP32 baseline, bringing OTA model updates into the feasible range for LoRaWAN Class C duty-cycle allocations.

TABLE II
EDGE DEPLOYMENT FEASIBILITY: QAT INT8 + PRUNED 70% VS. RASPBERRY PI 4B SOLAR BUOY CONSTRAINTS.

Constraint	Budget	Achieved	Pass
On-device RAM	512 MB	4.24 MB	✓
E2E Latency	<500 ms	336.88 ms	✓
Energy/Inference	<5 W active	0.0534 J	✓
OTA Payload	LoRaWAN viable	1.52 MB	✓

E. Pruning Sparsity and Latency Efficiency

The relationship between pruning sparsity and latency reduction confirms the memory-bandwidth bottleneck behavior modeled in Eq. (1). The 70% pruned variant reduces NZ-MACs from 310M to 93M (70.0% MAC reduction) but yields only a 9.1% E2E latency improvement relative to the FP32 baseline (370.4 ms to 336.88 ms). This plateau in speedup occurs because unstructured sparsity creates irregular memory access patterns. On the ARM Cortex-A72, the time saved by performing fewer calculations is offset by cache miss penalties, which become the primary bottleneck for workloads under 100M MACs. The primary latency gains instead are attributed to INT8 NEON SIMD vectorization, accounting for the the largest latency reduction between the FP32 baseline (370.4 ms) and the QAT INT8 model (346.79 ms).

F. Domain Shift Robustness

Evaluation under the three synthetic aquatic perturbation protocols demonstrates statistically negligible accuracy degradation for the QAT INT8 + Pruned 70% variant relative to the unperturbed validation baseline across all three modes—turbidity, biofouling, and sub-surface lighting variation—individually and in combination. The morphological feature representations encoded by the MobileNetV3-Small depthwise separable convolution stack are sufficiently distributed across the activation space to resist low-frequency spatial corruptions introduced by aquatic optical degradation. Critically, the 70%

pruned QAT INT8 variant exhibits no statistically significant sensitivity to domain perturbation relative to the FP32 baseline, indicating the pruned weight structure preserves similar distributional robustness properties.

V. DISCUSSION

A. Compression Pipeline Design Choices

The cascaded Deep Compression architecture reflects a deliberate stage ordering. QAT is applied prior to pruning because INT8 quantization boundaries must be established across the full dense weight manifold before connections are removed. Reversing this order compounds two independent sources of representational loss without the joint fine-tuning recovery mechanism that QAT affords. The joint QAT fine-tuning phase applied after each pruning threshold simultaneously compensates for missing connections and quantization noise within a single gradient descent trajectory, which is the cause of accuracy retention observed across the light and moderate compression structures.

CSR formatting is essential to the efficacy of the subsequent Huffman coding stage. While a dense INT8 tensor with 70% structural zeros possesses low theoretical entropy, Huffman coding alone cannot bypass the underlying memory allocation of the dense format. CSR formatting first physically removes these zero-valued entries, concentrating the Huffman input distribution onto non-zero INT8 discrete values. The combined effect produces the 74.6% total payload reduction in Table I, a result neither stage achieves in isolation.

B. Architecture Selection Rationale

The selection of MobileNetV3-Small over higher-capacity alternatives such as EfficientNet-B0 or Vision Transformer variants is governed by three factors. First, physical microplastic particles measuring ≤ 5 mm appear as severely low-resolution sub-pixel regions after 224×224 resizing; the spatial frequency content available for morphological discrimination is fundamentally limited by optical physics, not classifier capacity. Deploying a ViT to classify specks occupying < 100 active pixels burns energy while offering no marginal gain in extractable feature dimensionality. Second, MobileNetV3's depthwise separable structure is co-designed for ARM Cortex-A NEON SIMD execution. Third, 98.91% Top-1 accuracy empirically confirms that model capacity is not the binding constraint, validating the architecture selection.

C. The NOISE Class as a System-Level Design Decision

Prior vision-based microplastic detectors operate as binary detectors [3], delegating false-positive suppression to post-processing heuristics or human review. BuoyNet internalizes suppression within the classifier: any frame for which the model returns NOISE as the argmax prediction is silently discarded at the inference module level without generating a telemetry event. In a LoRaWAN deployment where each transmission consumes a non-trivial fraction of the daily duty-cycle allocation, suppressing false-positive transmissions at the inference layer directly extends operational battery life and

preserves bandwidth for genuine detection events. This choice has no accuracy cost as NOISE is a standard classification target during training and produces a system-level benefit not reflected in the Top-1 accuracy figures of Table I.

D. Simulation Validity and Limitations

The analytical latency model of Eq. (2) is grounded in Colab-measured `perf_counter` inference timing, published ARM Cortex-A72 NEON SIMD throughput ratios, and the established 0.55 W Raspberry Pi 4B active inference draw. This methodology is consistent with the mJ-per-image reporting convention established for Coral Dev Board deployments [10]. Three limitations must be acknowledged. First, the absence of physical hardware means TFLite layer fusion optimizations, OS scheduling jitter, and thermal throttling are not captured; physical deployment may produce latency variance of ± 10 –20% around simulated means. Second, all accuracy evaluations are performed on the shuffled validation set, introducing an optimistic bias relative to a fully held-out test evaluation. Third, domain shift perturbation protocols are synthetic and do not capture the full statistical distribution of optical degradation encountered in open-water Pacific deployment.

E. Comparison to Prior Work

BuoyNet advances the state of the art on two axes relative to the most comparable prior system. Plastiview [3] reports $\text{mAP@50} = 0.748$ on binary plastic detection with no compression pipeline and no energy reporting on identical Raspberry Pi 4B hardware. BuoyNet achieves 91.6% Top-1 accuracy on a 16-class morphological taxonomy, far more granular, at a 74.6% smaller transmission payload, with explicit energy-per-inference reporting. BuoyNet is, to the best of my knowledge, the first system in this domain to jointly report accuracy, compression ratio, transmission payload size, energy per inference, and domain shift robustness within a unified evaluation framework.

VI. CONCLUSION

This paper presented BuoyNet, an end-to-end edge machine learning pipeline for autonomous marine microplastic detection aboard a solar-powered IoT buoy node. A MobileNetV3-Small backbone fine-tuned via QAT achieves 98.91% Top-1 accuracy across 16 morphological classes, a classification granularity and accuracy ceiling not previously reported in vision-based microplastic detection research. A cascaded three-stage compression pipeline reduces the on-disk model from 5.98 MB to a 1.52 MB transmittable payload (74.6% reduction) while retaining 91.6% Top-1 accuracy at maximum compression. Hardware simulation for the Raspberry Pi 4B yields 336.88 ms E2E latency and 0.0534 J per inference, satisfying all four operational constraints of a 2 FPS solar-camera duty cycle. Robustness evaluation under synthetic aquatic domain perturbations confirms qualitatively minimal accuracy degradation, establishing that the pruned weight manifold preserves the distributional robustness properties of the dense FP32 network.

Three main limitations must be addressed before deployment: the lack of testing on physical hardware, the reliance on the validation dataset due to class imbalances, and the use of synthetic data rather than real-world underwater testing.

Four directions are identified for future work. First, the QAT model should be exported through ONNX to a TFLite FlatBuffer, enabling zero-idle-memory C++ runtime execution on Pi Zero 2W class hardware. Second, physical buoy deployment in a controlled reservoir environment would provide the first real-world aquatic microplastic classification dataset under real optical degradation conditions. Third, integration of multi-spectral LED illumination channels—ultraviolet excitation combined with white-light imaging—would enable fluorescence-based polymer species discrimination beyond the morphological taxonomy supported by the current RGB classifier. Fourth, GPS-tagged LoRaWAN payloads transmitted to a shore-side aggregation server enabling geospatial microplastic density mapping at ocean-basin scale.

BuoyNet demonstrates that cascaded deep compression, co-designed with the physical constraints of a solar IoT buoy platform, can bridge the gap between laboratory-grade microplastic characterization and scalable, real-time aquatic environmental monitoring. The implementation is publicly available at <https://github.com/Ishaan1402/BuoyNet>.

REFERENCES

- [1] T. Khanam et al., “Machine learning approaches for microplastic detection and classification: A comprehensive review,” *Frontiers in Environmental Science*, 2025.
- [2] L. Lin et al., “Raman hyperspectral imaging combined with machine learning for microplastic classification in aquatic environments,” *Chemosphere*, vol. 291, 2022.
- [3] Plastiview Development Team, “Plastiview: Real-time microplastic detection on embedded hardware using YOLOv5s,” 2024.
- [4] M. Hasan et al., “IoT-integrated image recognition system for microplastic detection and classification,” in *Proc. IEEE Conf.*, 2024.
- [5] M. A. B. Sarker, M. H. Imtiaz, T. M. Holsen, and A. B. M. Baki, “Real-time detection of microplastics using an AI camera,” *Sensors*, vol. 24, no. 13, p. 4394, 2024.
- [6] S. Han, H. Mao, and W. J. Dally, “Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2016.
- [7] M. Nagel et al., “Up or down? Adaptive rounding for post-training quantization,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021.
- [8] Y. Bengio, N. Léonard, and A. Courville, “Estimating or propagating gradients through stochastic neurons for conditional computation,” *arXiv preprint arXiv:1308.3432*, 2013.
- [9] A. Howard et al., “Searching for MobileNetV3,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019.
- [10] Google, “Coral Dev Board performance benchmarks,” Google AI Edge, <https://coral.ai/docs/dev-board/>, 2022.